Workshop Exercises — Crossover to Work Outcomes

Companion exercises for the exercise dyadic dataset (`data/exercise_data.RData`). This dataset is **independent** from the original Hahn-et-al-style dataset used by scripts 02–08. The exercise scenario studies how work-related psychological states of one partner (affect, self-determination, job crafting) crossover to influence the other partner's work outcomes (engagement, performance, creativity), and how three dyad-level moderators (live_together, years_together, time_spent_this_morning_together) shape those pathways.

Dataset summary

Element	Specification
N couples	250 (500 individuals)
Predictors (person-level)	`affect`, `sdt`, `job_crafting`
Outcomes (person-level)	`engagement`, `performance`, `creativity`
Crossover moderators (dyad-level)	`live_together` (binary), `years_together` (continuous), `time_spent_this_morning_together` (continuous)
Distinguishing variable	`gender` (male / female)
Within-dyad predictor correlations	~0.30 (affect, sdt, jc)
Within-dyad outcome correlations	~0.25 (engagement, performance, creativity)
Partner predictors in long format	`partner_affect`, `partner_sdt`, `partner_job_crafting`
Wide format	`_a` / `_p` suffixes (actor / partner)

Focal moderated-crossover effects (one per outcome)

Outcome	Focal partner predictor	Focal moderator	DGP interaction
engagement	`partner_job_crafting`	`live_together`	partner_jc × live_together = 0.15
performance	`partner_sdt`	`years_together`	partner_sdt × years_together = 0.015

Outcome	Focal partner predictor	Focal moderator	DGP interaction
creativity	`partner_affect`	`time_spent_this_morning_together`	partner_affect × time_spent = 0.003

The DGP also includes one gender-by-actor effect: `actor_sdt × gender_male` on `engagement` equals 0.10. All other slopes are equal across gender.

Per-section outcome

Each section focuses on **one** outcome so the multi-outcome repetition does not dominate the time budget. Every outcome still appears at least twice across the set. Section 1 inspects all three. Section 9 lets you pick.

Section	Outcome
1. Data simulation & inspection	all three (ICC, t-tests, summaries)
2. Indistinguishable dyads: MLM	`engagement`
3. Indistinguishable dyads: SEM	`creativity`
4. Distinguishable dyads: MLM moderator	`engagement`
5. Distinguishable dyads: SEM (long)	`performance`
6. Distinguishable dyads: SEM wide	`engagement`
7. Two-intercept models	`creativity`
8. Moderated APIM (centrepiece)	`engagement`
9. Integrative exercise	your choice

Files

File	Purpose
`exercises/simulate_exercise_data.R`	DGP script (run once to regenerate `data/exercise_data.RData`)
`data/exercise_data.RData`	Contains `ddl2` (long) and `ddw2` (wide)
`exercises/exercises.md`	This file
`exercises/answers_template.R`	Step-by-step instructions and code stubs
`exercises/completed_exercises.R`	Fully worked solutions matching this file

How to use these exercises

The exercises assume you have the exercise dataset loaded and the workshop's reference scripts (02–08) available for syntax. Each section below describes the goal and what to produce. Open `exercises/answers_template.R` for the step-by-step instructions and code stubs. After producing each output, write a one-sentence interpretation before moving on — that articulation is the exercise.

Reference script numbers below refer to `scripts/02_indistinguishable_mlm.R` through `scripts/08_moderated_apim.R`. Section 5 uses `scripts/06_distinguishable_sem_longformat.R` and Section 6 uses `scripts/07_distinguishable_sem_wide.R` — the per-section "Reference" line in the template makes the mapping explicit.

Section 1 — Data simulation & inspection

What you are practising: orienting yourself to a new dyadic dataset before any modelling.

Reference: `exercises/simulate_exercise_data.R` (the DGP) and `scripts/02_indistinguishable_mlm.R` (for the `icc()` helper).

What to produce: Read the header of the simulation script to understand the data-generating parameters for the three outcomes and the focal moderated-crossover effects. Run the script to create the data file. In a fresh R session, load the data and inspect both the long and wide formats. For each of the three outcomes, fit a null multilevel model and compute the ICC to assess within-dyad dependence. Summarise the three dyad-level moderators to understand their distributions. Run paired t-tests on the three outcomes to identify which show non-zero gender mean differences.

Reflection prompt: Which outcome shows the largest gender difference in mean? Does it match the DGP intercept gap, or is it diluted by the gender-asymmetric predictors?

Section 2 — Indistinguishable dyads: multilevel modelling

Outcome: `engagement`.

What you are practising: the basic Actor-Partner Interdependence Model in long format, treating dyad members as exchangeable.

Reference: `scripts/02_indistinguishable_mlm.R`.

What to produce: Load the long-format data. Fit the full APIM for `engagement` with all three actor predictors, all three partner predictors, the three dyad-level moderators, and a random intercept for dyad. Fit a reduced model that omits the three partner predictors, and use a likelihood ratio test to check whether the partner block is jointly needed. Compute profile-likelihood confidence intervals and write a two-sentence interpretation of the actor and partner effects.

Reflection prompt: Is the partner block jointly significant? Does the answer match the DGP, which has non-zero partner effects for all three predictors on `engagement`?

Section 3 — Indistinguishable dyads: SEM

Outcome: `creativity`.

What you are practising: three structural equation modelling paradigms for indistinguishable dyads and the formal indistinguishability test.

Reference: `scripts/03_indistinguishable_sem.R`.

What to produce: Load the wide-format data. For `creativity`, fit three structural equation models: a cluster-robust model in long format, a two-level model that decomposes effects into within-dyad and between-dyad components, and a wide-format model with full equality constraints across dyad members. Then fit an unconstrained wide-format model and use a likelihood ratio test to assess whether the equality constraints are tenable. The DGP has equal slopes on `creativity` across gender, so the test should fail to reject the indistinguishability null.

Reflection prompt: Does the likelihood ratio test confirm indistinguishability for `creativity`? What does that imply about the `actor_sdt × gender_male` interaction — is it on `engagement` only, or could it also appear here?

Section 4 — Distinguishable dyads: MLM moderator

Outcome: `engagement`.

What you are practising: adding `gender` as a moderator of actor and partner effects in a multilevel model.

Reference: `scripts/04_distinguishable_mlm.R`.

What to produce: Load the long-format data. Fit a baseline multilevel model for `engagement` (no gender interactions) and a full moderator model that interacts gender with every actor and partner predictor. Compare the two via a likelihood ratio test. Use simple slopes to extract the `actor_sdt` effect separately for each gender. The DGP has only one non-zero gender interaction: `actor_sdt × gender_male` on `engagement`.

Reflection prompt: Does the simple-slope plot for `actor_sdt` by gender show a clear male-female difference, in line with the DGP, or do other gender interactions appear spuriously significant?

Section 5 — Distinguishable dyads: SEM (two-level with moderation)

Outcome: `performance`.

What you are practising: mirroring the multilevel moderator approach in SEM with calculated simple slopes.

Reference: `scripts/06_distinguishable_sem_longformat.R`.

What to produce: Load the long-format data. Construct a numeric gender indicator and create manual interaction columns for every actor and partner predictor. Specify a two-level SEM for `performance` with the gender interactions at the within-dyad level and the three moderators at the between-dyad level. Add calculated simple-slope parameters (one per gender per pathway) using the lavaan `:=` operator. Fit the model and compare the within-dyad SEM gender coefficients with the multilevel estimates from a simple re-fit on `performance`. The DGP has no gender interactions on `performance`, so the gender block should be small.

Reflection prompt: Do the SEM gender coefficients agree with a quick MLM re-fit on `performance`? Where do they differ, and is the difference substantively meaningful?

Section 6 — Distinguishable dyads: SEM wide format

Outcome: `engagement`.

What you are practising: the unconstrained / slopes-equal / fully-constrained model sequence and the three nested likelihood ratio tests.

Reference: `scripts/07_distinguishable_sem_wide.R`.

What to produce: Load the wide-format data. For `engagement`, fit three wide-format SEMs: one with all paths free to vary across dyad members, one that constrains slopes to be equal but leaves intercepts free, and one that constrains everything (slopes, intercepts, residual variances). Run the three nested likelihood ratio tests per outcome: slope equality, intercept equality, and overall indistinguishability. Report each p-value. The DGP has equal slopes on `engagement` (the only gender interaction is `actor_sdt × gender_male`, which the wide-format unconstrained vs slopes-equal test may flag) and a non-zero intercept gap.

Reflection prompt: Which of the three LRTs is most decisive? Does the slope-equality test pick up the `actor_sdt × gender_male` DGP effect in the wide-format SEM?

Section 7 — Two-intercept models

Outcome: `creativity`.

What you are practising: the parameterisation trick that directly estimates separate intercepts per group, and its SEM analogue.

Reference: `scripts/05_two_intercept_models.R`.

What to produce: Load the long-format data. For `creativity`, fit three nested multilevel models: one with equal intercepts and equal slopes, one with separate intercepts for each gender but equal slopes, and one with separate intercepts and separate slopes. Use the parameterisation that suppresses the global intercept and replaces it with separate group intercepts (`0 + gender` in `lmer`). Compare adjacent models with likelihood ratio tests. Then fit the SEM analogue in wide format (two intercepts, equal slopes) and confirm the intercept estimates match the multilevel model. The DGP has a negative male-vs-female intercept gap on `creativity` (males 4.80, females 4.95), so the two-intercept approach should reveal a clear mean reversal.

Reflection prompt: The two-intercept approach is recommended when the researcher wants to report the absolute mean of each group. Does the `creativity` model show a substantively interesting mean reversal between males and females?

Section 8 — Moderated APIM (the centrepiece)

Outcome: `engagement`.

What you are practising: testing whether the partner crossover effect is moderated by a dyad-level variable, using both multilevel modelling and SEM.

Reference: `scripts/08_moderated_apim.R`.

What to produce: Load both the long- and wide-format data. For `engagement`, fit a baseline multilevel model and a moderated multilevel model that includes the focal `partner_job_crafting × live_together` interaction (DGP value 0.15). Use simple slopes to extract the `partner_job_crafting` effect at the two levels of `live_together`, and plot the interaction. Run the likelihood ratio test comparing base to moderated. Then build the wide-format SEM with a manually created product indicator and computed simple-slope parameters, and compare the simple-slope estimates with those from the multilevel model.

Reflection prompt: Do the multilevel and SEM simple-slope estimates agree? Where do they differ, and is the difference larger for the binary moderator (`live_together`) than you would expect for a continuous moderator?

Section 9 — Integrative exercise

Outcome: your choice (one of the three).

What you are practising: combining everything you have learned to answer a single, fully-specified research question.

What to produce: Pick one outcome (e.g. `engagement`, since it has the richest DGP). Build a kitchen-sink multilevel model that includes all three actor predictors, all three partner predictors, all three dyad-level moderators, the focal crossover-by-moderator interaction for your chosen outcome, and a random intercept. Build the corresponding wide-format SEM with a manually created product indicator and computed simple-slope parameters at three levels of the moderator (low, mean, high; for the binary `live_together`, use 0 and 1). Report the actor and partner effect estimates with standard errors, the focal interaction coefficient from both approaches, the three simple slopes, and the SEM fit indices. Write a 4–6 sentence interpretation paragraph covering the substantive size and direction of the partner crossover effect, whether the moderator meaningfully changes that effect, a plain-language interpretation, and one limitation or follow-up question.

Reflection prompt: If you had to present this in a 5-minute conference talk, which figure and which one-sentence result would you lead with?